10

four different chemical languages

IUPAC Name:

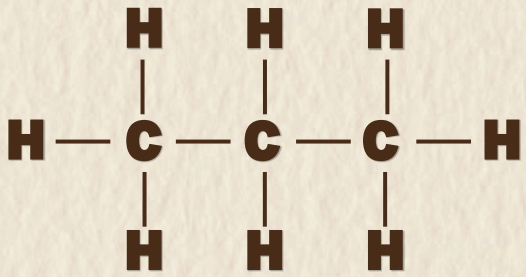
PROPANE

Propane is a liquefied petroleum gas that comes out of both oil and gas wells.

MOLECULAR FORMULA

C_3H_8

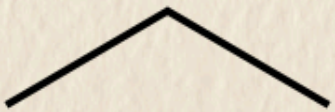
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name:

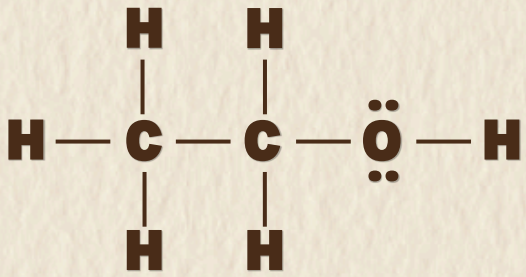
ETHANOL

Ethanol, also known as *ethyl alcohol* or simply alcohol, is a chemical compound that is both colorless and flammable.

MOLECULAR FORMULA

C_2H_6O

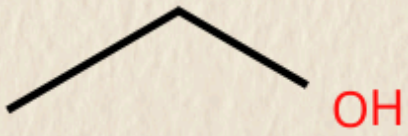
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **Propan-2-one**

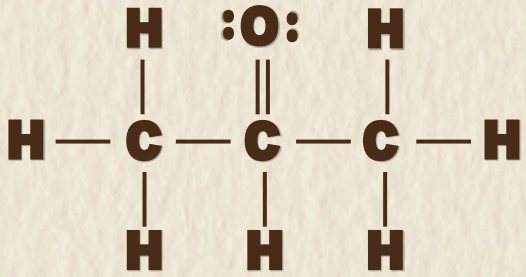
ACETONE (PROPAN-2-ONE)

Acetone is a liquid solvent that can break down and dissolve other substances.

MOLECULAR FORMULA

C_3H_6O

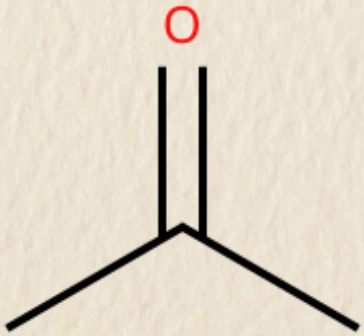
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3COCH_3

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **Ethanoic acid**

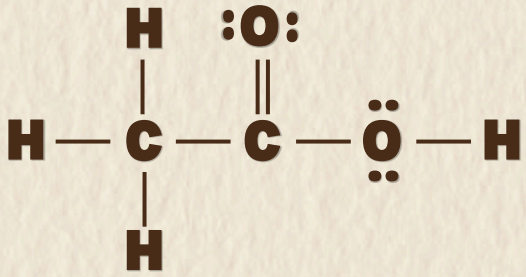
ACETIC ACID (ETHANOIC ACID)

Acetic acid, also known as ethanoic acid, is a chemical compound found in many different products. It's perhaps most well-known as the main component of vinegar.

MOLECULAR FORMULA

$C_2H_4O_2$

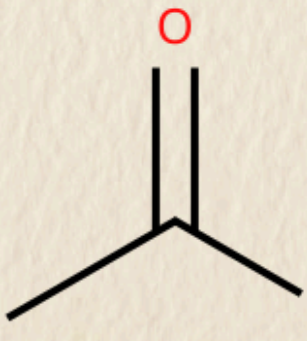
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3COOH

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **Ethyl ethanoate**

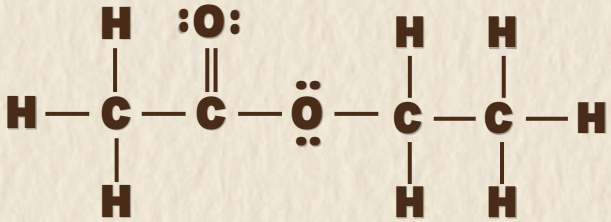
ETHYL ACETATE (ETHYL ETHANOATE)

Ethyl Acetate is an organic ester compound that appears as a colorless liquid. It is highly miscible with all common organic solvents.

MOLECULAR FORMULA

$C_4H_8O_2$

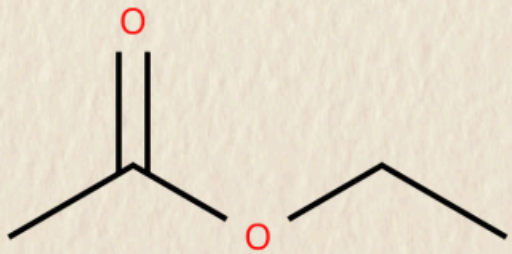
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3COOC_2H_5$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

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four different chemical languages

IUPAC Name:

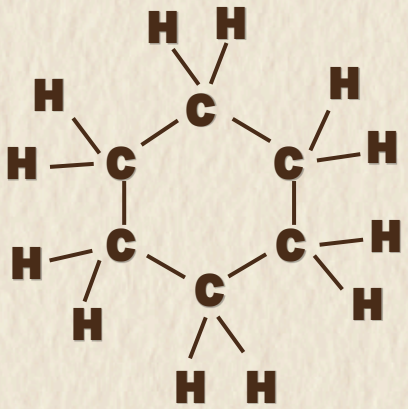
CYCLOHEXANE

Cyclohexane appears as a clear colorless liquid with a petroleum-like odor. Used to make nylon, as a solvent, paint remover, and to make other chemicals.

MOLECULAR FORMULA

C_6H_{12}

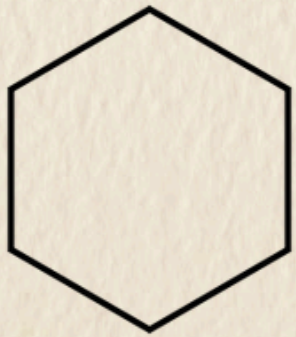
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$(CH_2)_6$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

07

four different chemical languages

IUPAC Name:

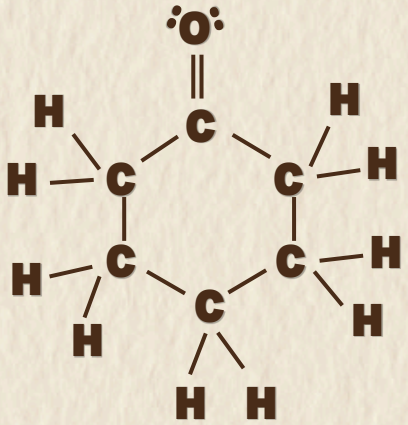
CYCLOHEXANONE

Cyclohexanone is an organic compound that is a colorless to pale yellow liquid with an odor similar to that of acetone that is very soluble in most organic solvents.

MOLECULAR FORMULA

$C_6H_{10}O$

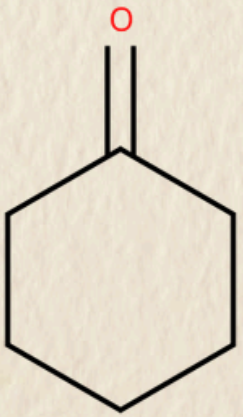
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$(CH_2)_5CO$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **N-(4-hydroxyphenyl)acetamide**

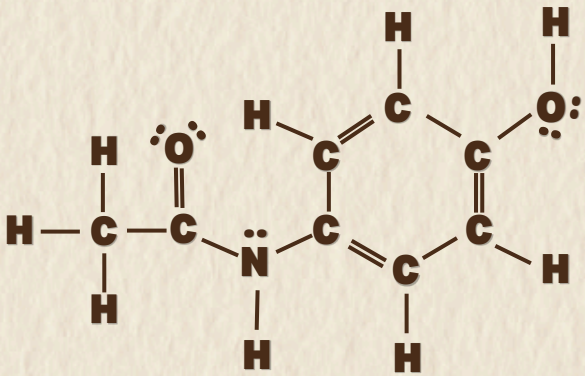
ACETAMINOPHEN (PARACETAMOL)

Acetaminophen (paracetamol), also commonly known as *Tylenol*, is the most commonly taken analgesic worldwide and is recommended as first-line therapy in pain conditions by the World Health Organization (WHO).

MOLECULAR FORMULA

$C_8H_9NO_2$

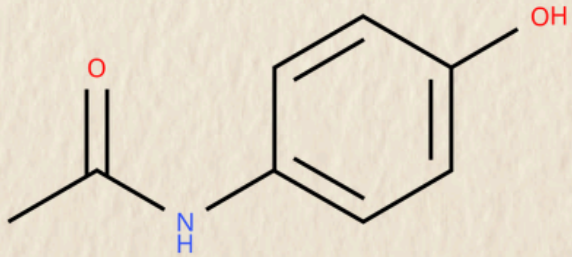
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$p\text{-}HOC_6H_4NHCOCH_3$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **2-(Acetyloxy)benzoic acid**

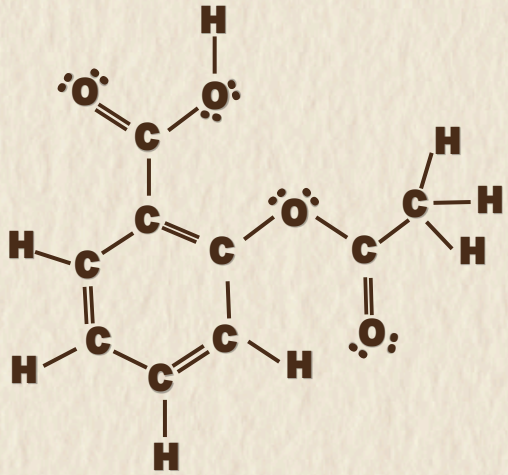
ASPIRIN (ACETYLSALICYLIC ACID)

Aspirin (acetylsalicylic acid) is one of the oldest and most commonly used non-steroidal anti-inflammatory drugs (NSAIDs) for treating mild-to-moderate pain, fever, and inflammation from various causes.

MOLECULAR FORMULA

$C_9H_8O_4$

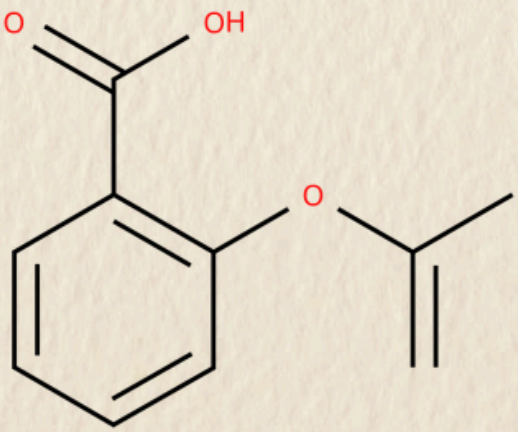
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3COOC_6H_4COOH$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

IUPAC Name: **1,3,7-Trimethylpurine-2,6-dione**

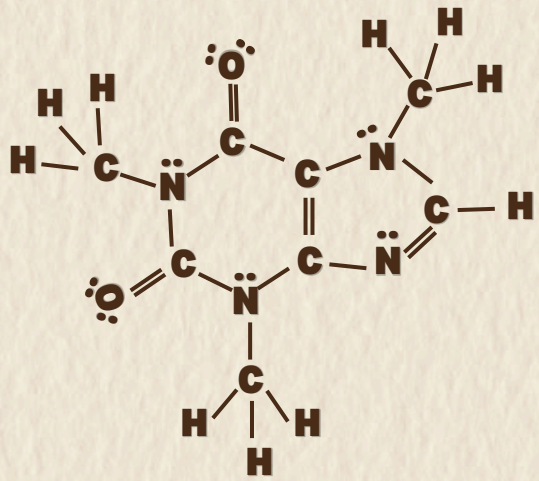
CAFFEINE

Caffeine is a drug of the methylxanthine class used for a variety of purposes, including certain respiratory conditions of the premature newborn, pain relief, and to combat drowsiness.

MOLECULAR FORMULA

$C_8H_{10}N_4O_2$

LEWIS STRUCTRE



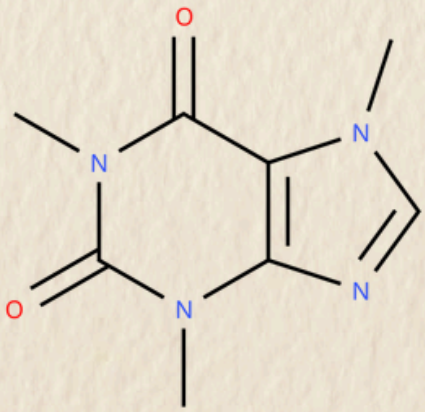
CONDENSED STRUCTURAL FORMULA

Caffeine does not have a simple, widely accepted condensed structural formula. Therefore, its structure may be represented using a structural diagram or SMILES notation. SMILES is a notation used to describe a molecule's structure and is not the same as a condensed structural formula.

SMILES Notation:

CN1C=NC2=C1C(=O)N(C(=O)N2C)C

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name